Fast LLM-watermark segmentation

(And insights from textGrain)

Soham Bonnerjee

Postdoc, Wharton Stats

Based on joint research with

Paper 1



Sayar Karmakar



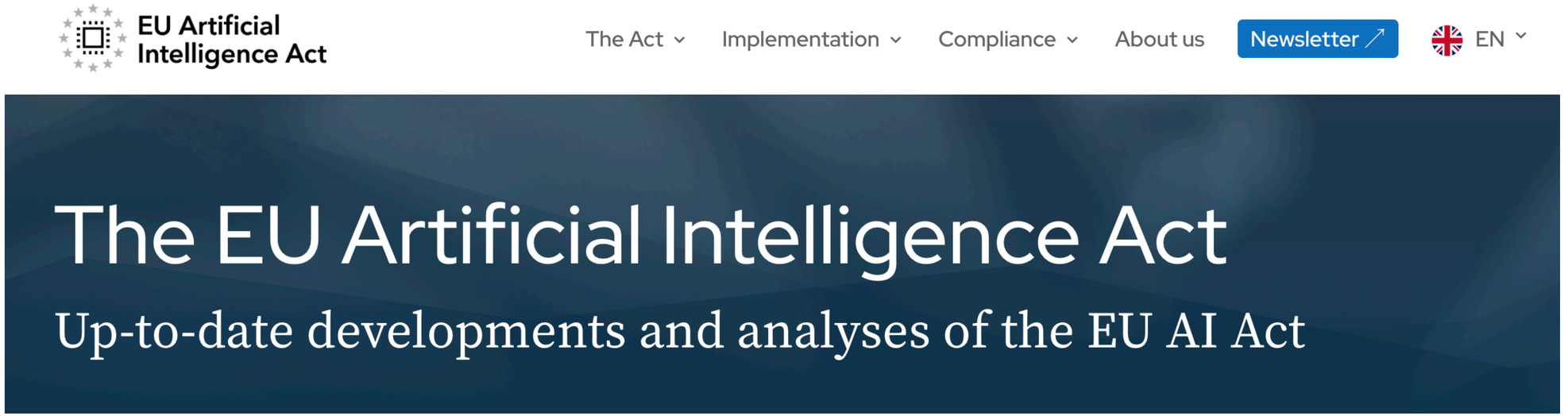
Subhrajyoti Roy

Paper 2



Swapnaneel Bhattacharyya

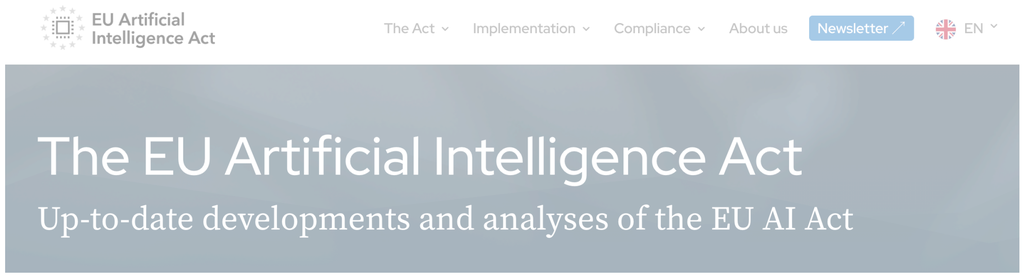
LLM Watermarking is very common



"LLM-generated texts need machine-readable watermarks"

Also: US President Executive Order 14110

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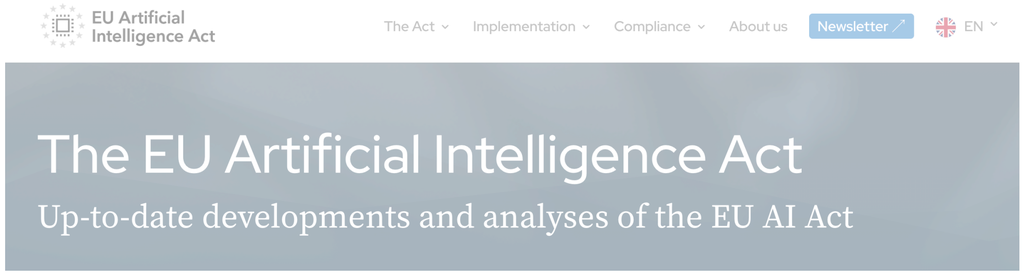
textGrain: Entropy-Calibrated Watermarking for Language Model Text

Technical report accompanying OpenAI's October 5th blog post

Xiang Li, University of Pennsylvania
Garrett Wen, Yale University
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October 5, 2026

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How Claude marks content

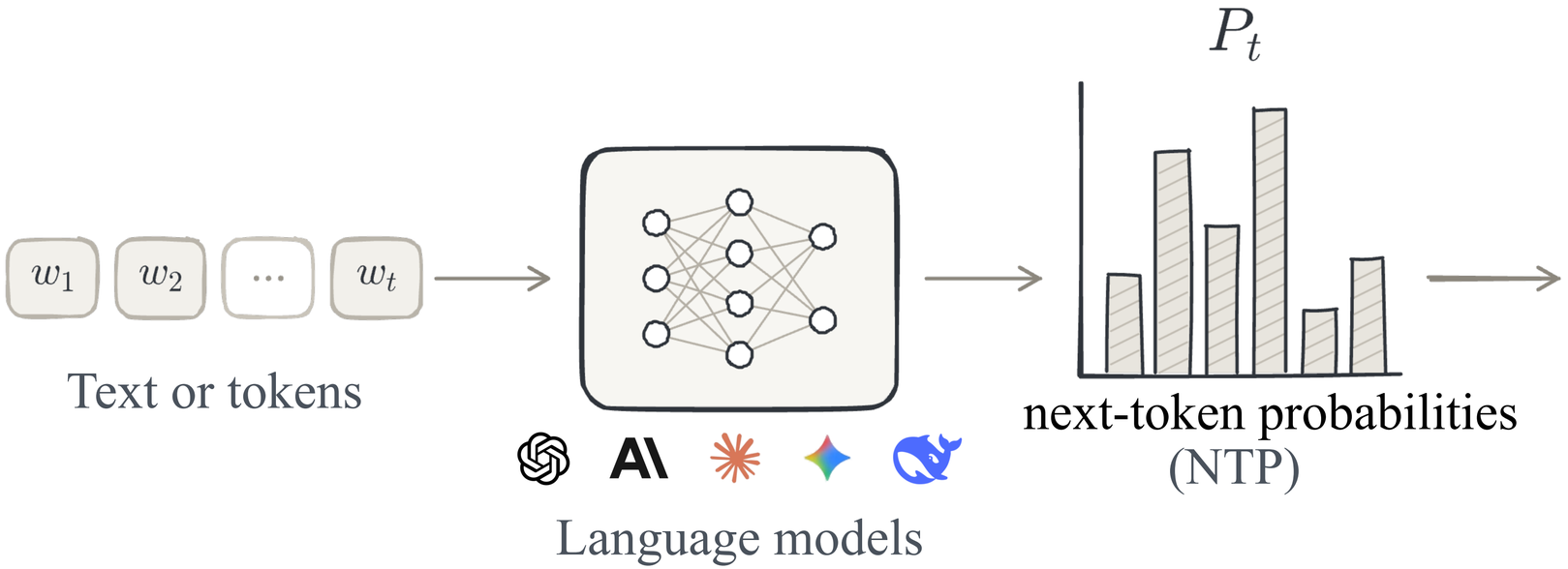
Claude uses two complementary techniques to mark content generated and processed by Claude: (1) watermarks embedded in text, and (2) signed provenance metadata attached to files.

1. Embedded watermarks in text

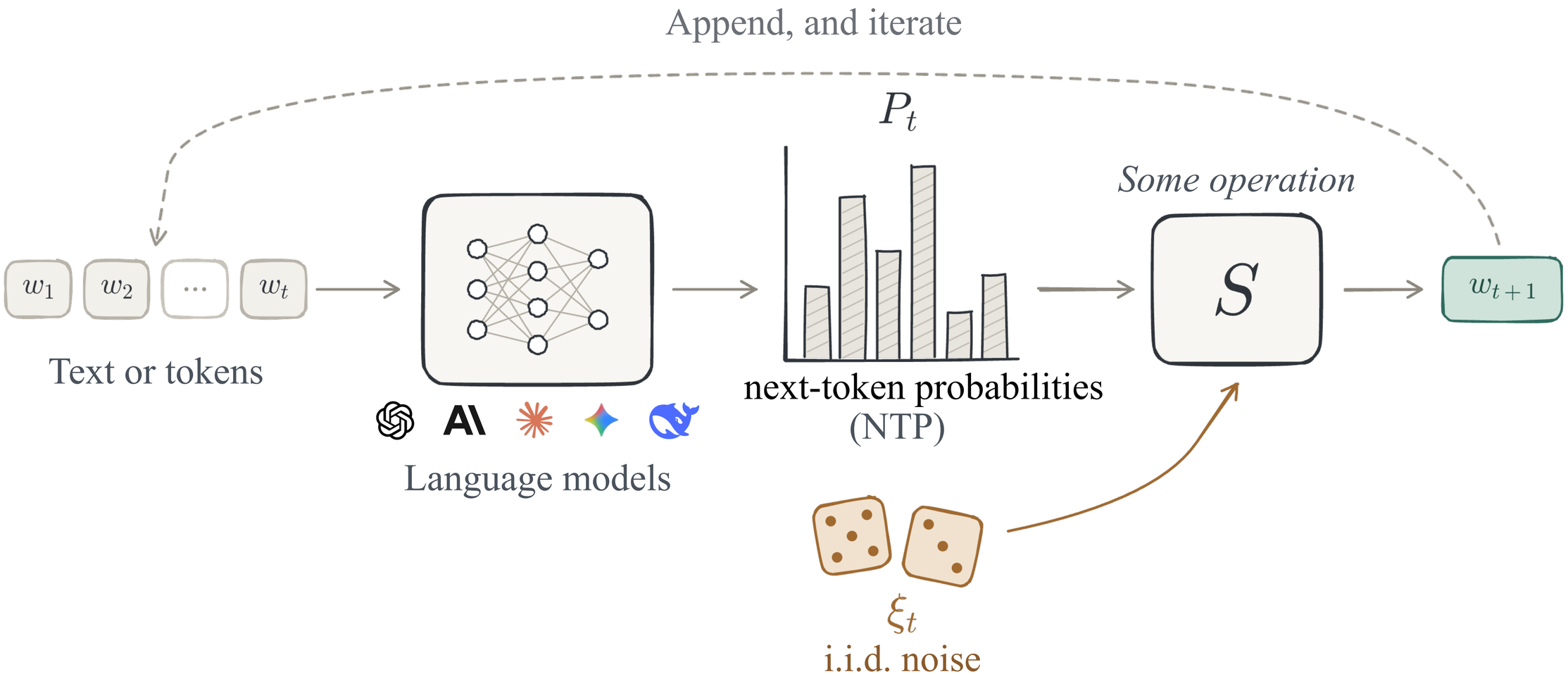
When a supported Claude model generates text, it weaves an imperceptible watermark directly into the text itself. You won't see it, and it doesn't change the meaning, quality, or readability of Claude's response.

Because the watermark is part of the text, it will travel with the text when it's copied and pasted elsewhere, and may persist through some editing. Watermarking will be applied at the model level, which means it will be present no matter which Claude product or surface the text comes from.

How does LLM Watermarking (encoding) work?



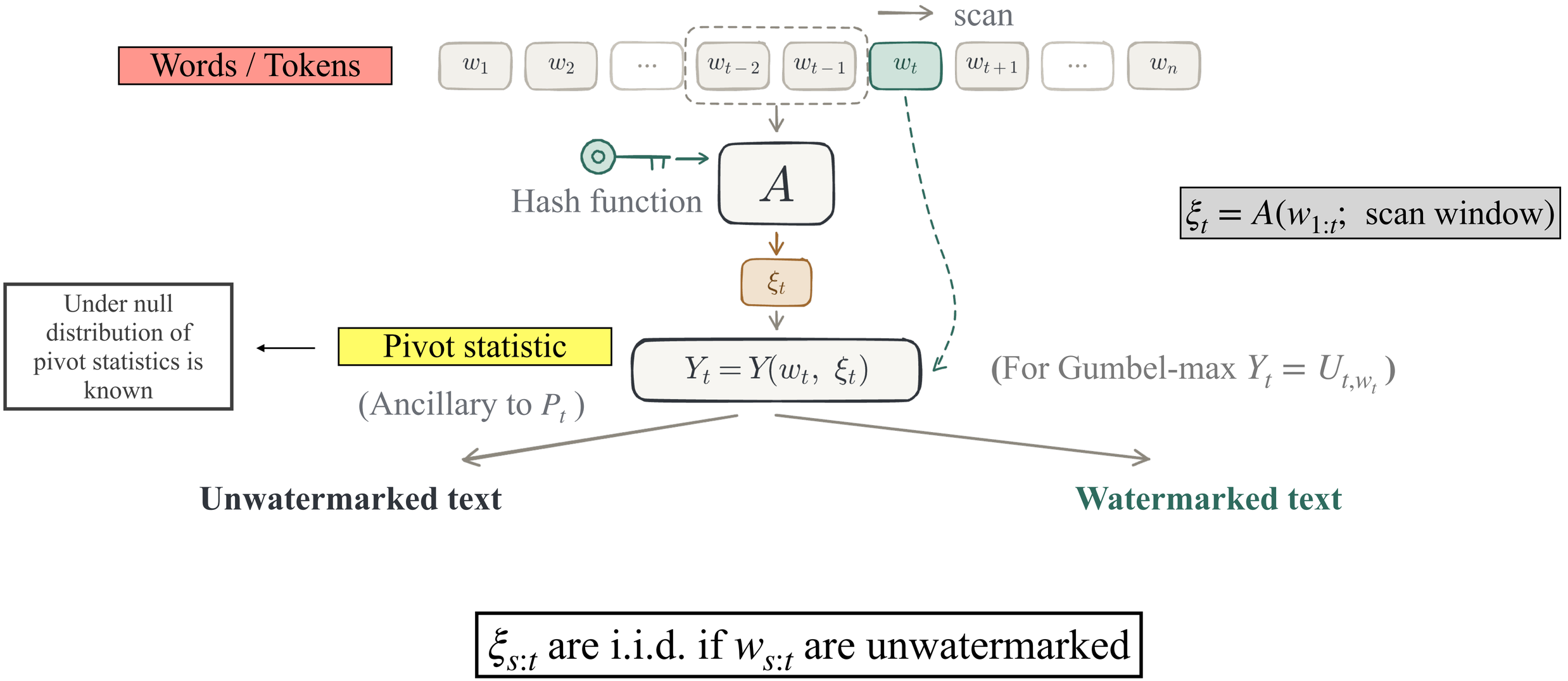
How does LLM Watermarking (encoding) work?



$$\boxed{w_{t+1} = S(P_t,\ \xi_t)}$$

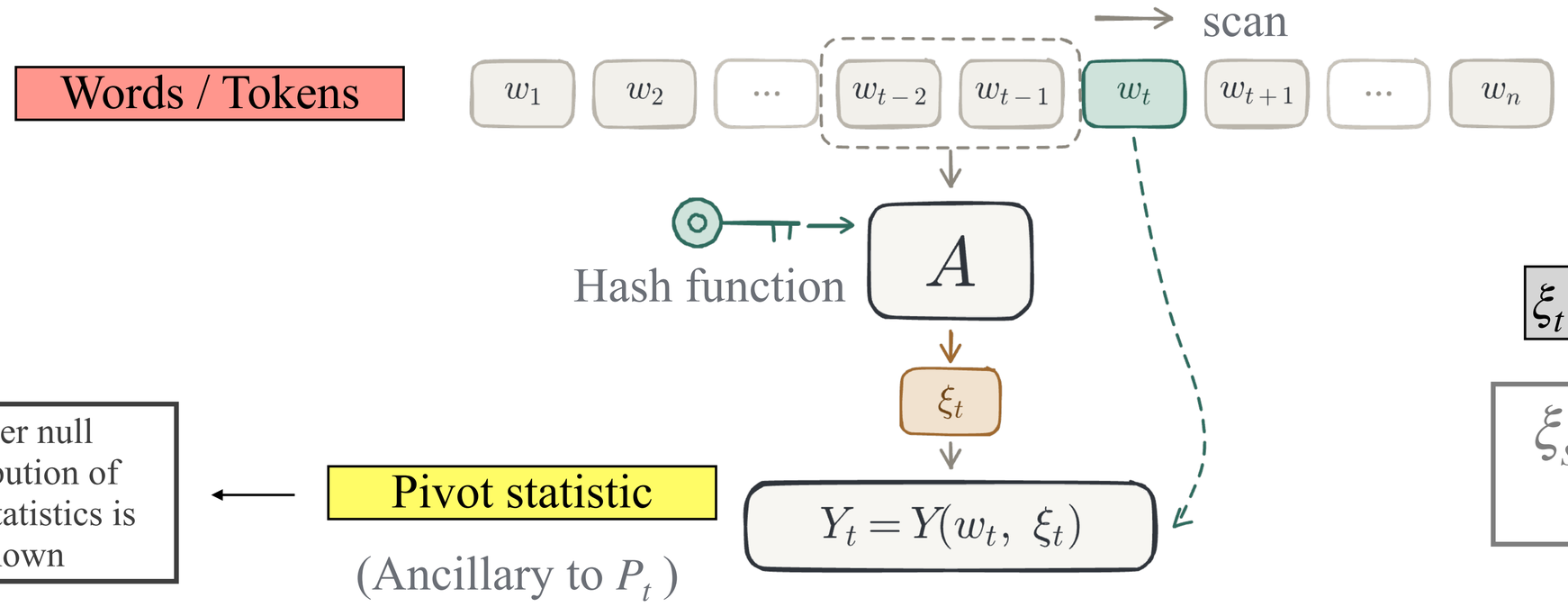
Eg. Gumbel-max: $w_{t+1} = \arg\max_w \dfrac{\log \xi_{t,w}}{P_{t,w}},\ \xi_{t,w} \overset{i.i.d.}{\sim} U[0,1]\,.$

How does watermark detector (decoding) work?



$\xi_{s:t}$ are i.i.d. if $w_{s:t}$ are unwatermarked

How does watermark detector (decoding) work?



$$\xi_t = A(w_{1:t};\ \text{scan window})$$

$\xi_{s:t}$ are i.i.d. if $w_{s:t}$ are unwatermarked

Under null distribution of pivot statistics is known

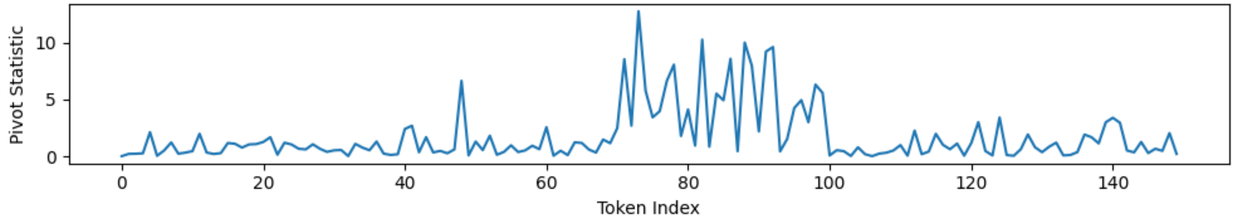
← **Pivot statistic**

(Ancillary to P_t)

Legend: Unwatermarked Gumbel Watermark

It is possible more people who came in contact with a man who died from the Ebola virus in Dallas, Texas, could test positive for the virus in the coming days, the head of the Centers for Disease Control and Prevention said Sunday. Tomohiro Mori, director of the CFDA'S Washington state office, confirmed to CNN that three cases of Ebola North American have been confirmed in the Dallas-area. Two of thoseechoe since 24 Saturday were transferred to Democratic Texas state hospitals because of random sneezing, said Margaret Summers of the Farrand Health System in Washington. One patient who traveled to New York âĢĶ a man from Japan whose travel had presumably been made by the Ebola vaccine but who traveled all from Japan

Pivot statistic vs token index



Two key questions

- Detection (Testing): Does this document contain LLM watermarking?
  - Cai et al. 2024, Kuditipudi et al. 2024, Li et al. 2025, Cai et al. 2025.
  - Analysis of textGrain, OpenAI's newest watermarking which preserves entropy of human-generated texts
    Does entropy preservation make detection harder?
    Bhattacharyya and **B.** (2026)

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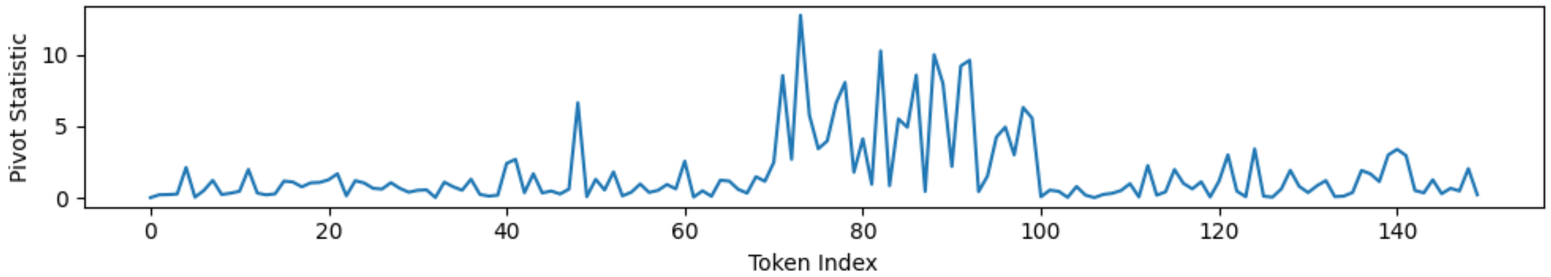
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  - WISER: a fast, statistically valid watermark segmentation algorithm[1]. **B.** et al. (2026+)

[1] Follow-up other recent localization algorithms: Blanchet et al. (2026), Li et al. (2026)

Watermark segmentation is epidemic localization

Legend: Unwatermarked Gumbel Watermark

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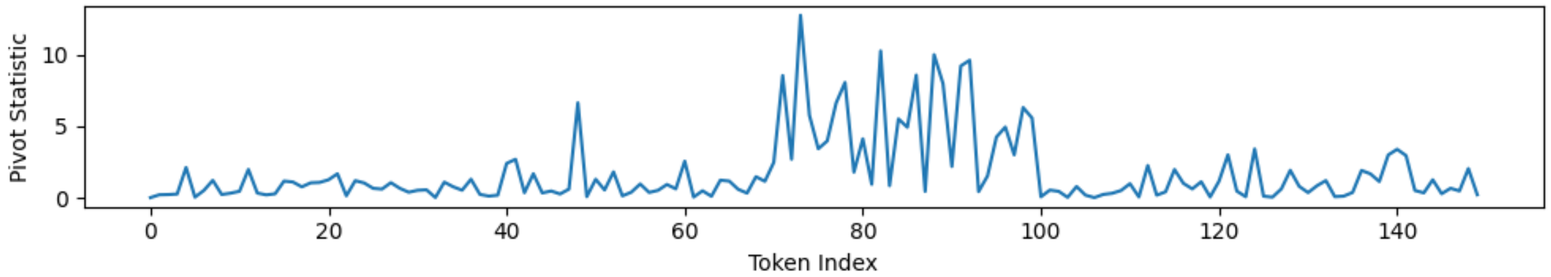


- Existing theory is for i.i.d./stationary.
- Dependence between watermarked tokens is very non-stationary.

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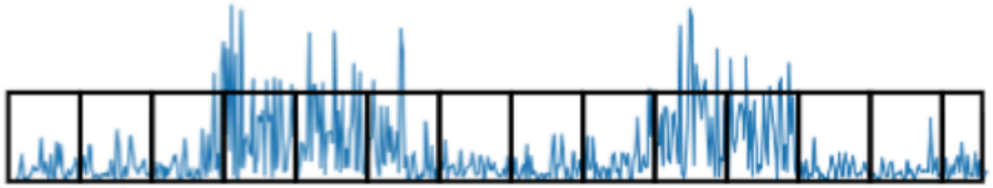
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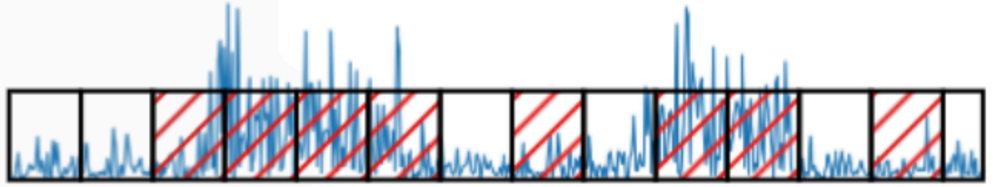
- Existing theory is for i.i.d./stationary.
- Dependence between watermarked tokens is very non-stationary.
- **We estimate number of patches, and then estimate each patch.**

WISER: a coarse-to-fine approach

Pivot statistic vs token index



Take blocks of size b and calculate the block sums.

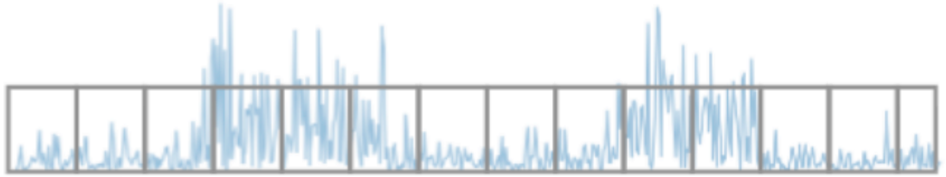


The block sums of red shaded blocks exceed Q

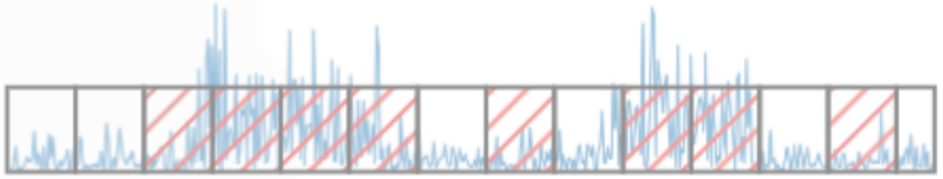
To determine Q, take (1-α)-th quantile of null distribution of maximum block sums → Under null distribution of pivot statistics is known

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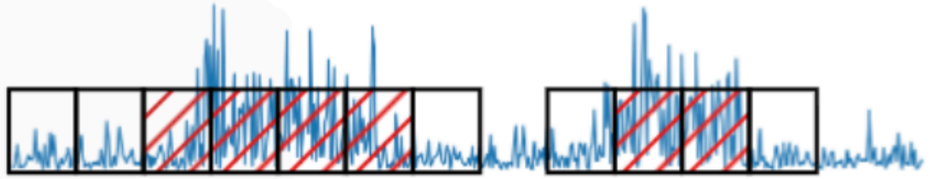


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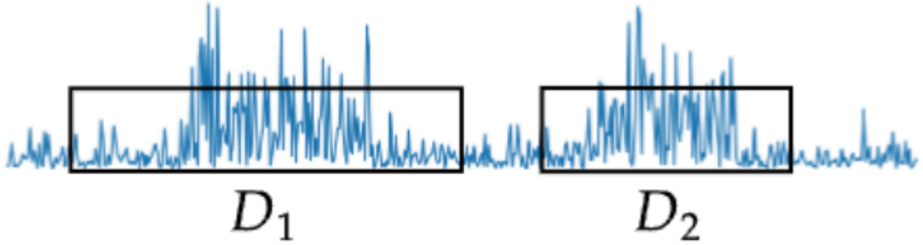


The block sums of red shaded blocks exceed $\mathbb{Q}$

To determine Q, take (1-α)-th quantile of null distribution of maximum block sums



Form connected intervals based on selected blocks.
Remove the connected intervals with relatively small lengths.



Slightly enlarge the remaining connected intervals to form intervals D_i.

Watermark Identification via Segmenting Epidemic Region

(a linear-time coarse-to-fine approach)

- Inside each interval D_j :

Null mean

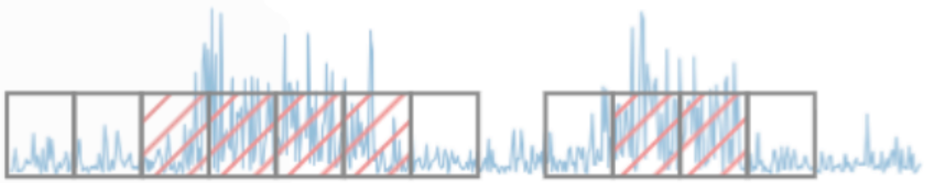
$$\hat{I} = \arg\min_{s,t \in D_j} \sum_{k \notin [s,t] \setminus D_j} (Y_k - \mu_0 - \rho\,\tilde{d}),\ \rho \in (0,1)\,.$$

$\tilde{d} \approx$ sharpest d, estimated from data.

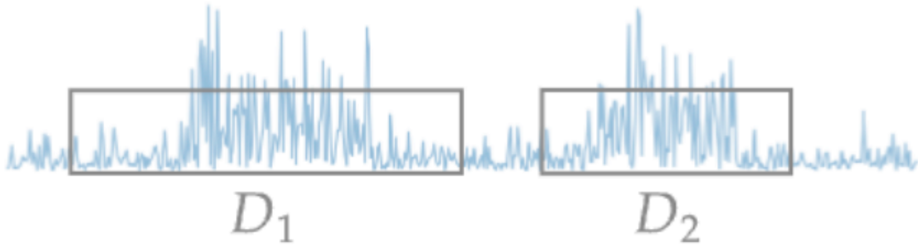
Under null distribution of pivot statistics is known

Assumption
(holds for all popular watermarking schemes)

$$\inf_P \mathbb{E}_{P,\ \text{watermark}}(Y_t) \geq \mu_0 + d$$



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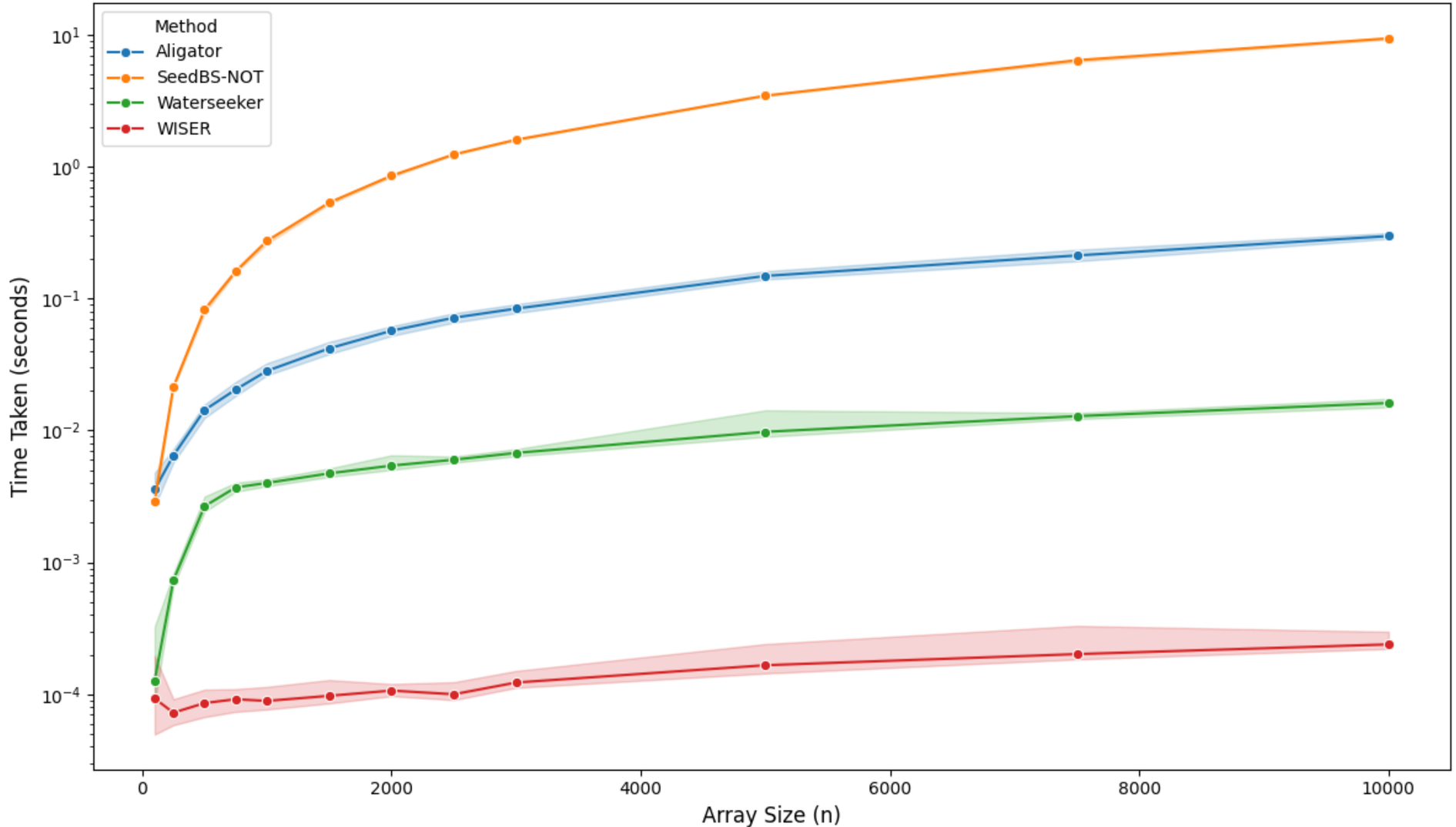


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to form intervals D_i.

WISER is fast

Time taken to identify a random $n/6$-length watermarked block in a text of n words.



WISER is accurate

Theorem (informal)

Under regularity conditions

+

Minimal separation between watermarked blocks,

$$\mathbb{P}\left(\hat{K} = K;\ \min_{1 \le j \le K} |I_j \Delta \hat{I}_j| < \frac{M}{d}\right) \to 1.$$

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Oracle uses the knowledge of the number of watermarked Patches

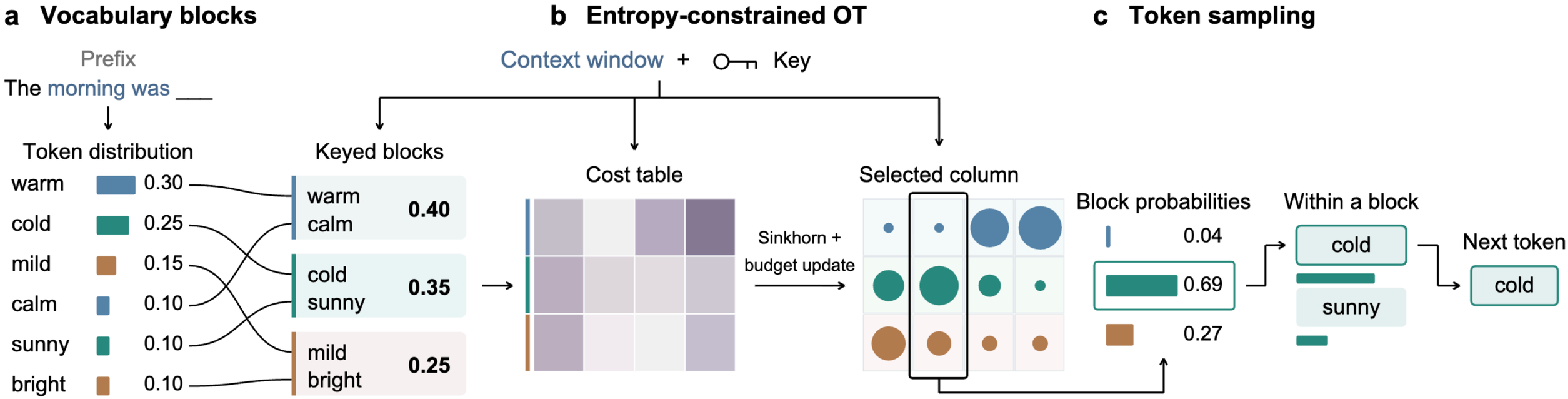
google gemma-3-270m (262144)	Oracle	0.916 (0.14-0.99)	0.980	0.980	0.980	0.976 (0.75-1.00)
	WISER	**0.903 (0.00-0.99)**	0.965	**0.960**	**0.962**	**0.953 (0.20-1.00)**
	Aligator	0.539 (0.00-0.87)	0.252	0.920	0.395	0.881 (0.20-0.97)
	SeedBS	0.299 (0.00-0.86)	0.562	0.420	0.481	0.727 (0.20-0.96)
	WaterSeeker	0.646 (0.12-0.93)	**0.973**	0.780	0.866	0.854 (0.69-0.98)
meta-llama Meta-Llama-3-8B (128256)	Oracle	0.940 (0.74-0.99)	0.995	0.995	0.995	0.982 (0.94-1.00)
	WISER	**0.928 (0.69-0.99)**	**1.000**	**0.988**	**0.994**	**0.977 (0.91-1.00)**
	Aligator	0.572 (0.01-0.92)	0.379	0.940	0.540	0.915 (0.71-0.98)
	SeedBS	0.383 (0.00-0.98)	0.645	0.510	0.570	0.768 (0.20-0.99)
	WaterSeeker	0.570 (0.33-0.91)	1.000	0.720	0.837	0.814 (0.67-0.98)
mistralai Mistral-7B-v0.1 (32000)	Oracle	0.920 (0.68-0.99)	0.998	0.998	0.998	0.978 (0.92-1.00)
	WISER	**0.880 (0.59-0.98)**	**1.000**	**0.998**	**0.999**	**0.965 (0.87-1.00)**
	Aligator	0.310 (0.00-0.86)	0.226	0.750	0.348	0.823 (0.55-0.96)
	SeedBS	0.220 (0.00-0.79)	0.405	0.278	0.329	0.611 (0.20-0.95)
	WaterSeeker	0.621 (0.36-0.88)	1.000	0.765	0.867	0.840 (0.69-0.97)

A brief foray into textGrain

OpenAI’s latest (Oct 5, 2026) watermark attempts to maintain entropy of unwatermarked text

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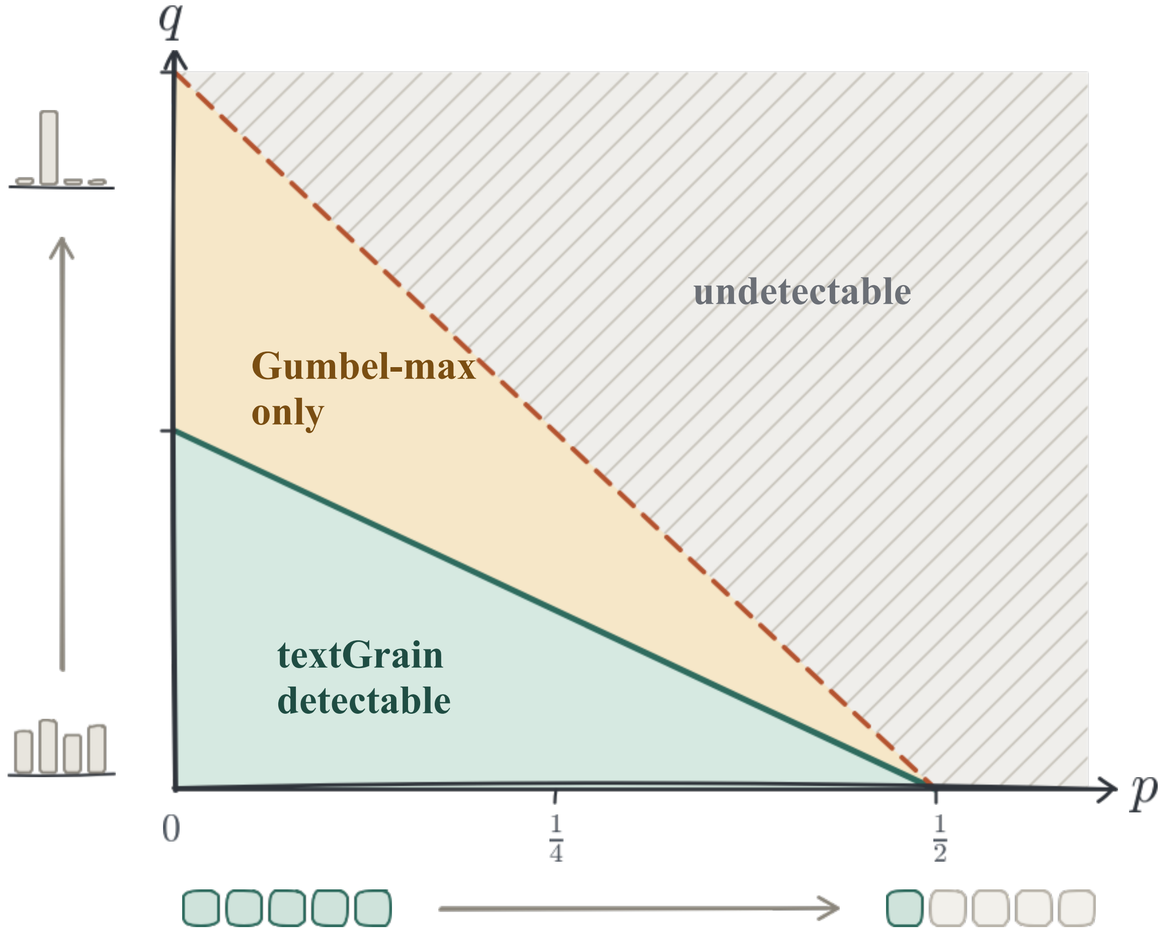
Schematic from Li et al. (2026).

Detection boundaries for textGrain

Bhattacharyya and **B.** (2026)

more predictable tokens; harder to detect.

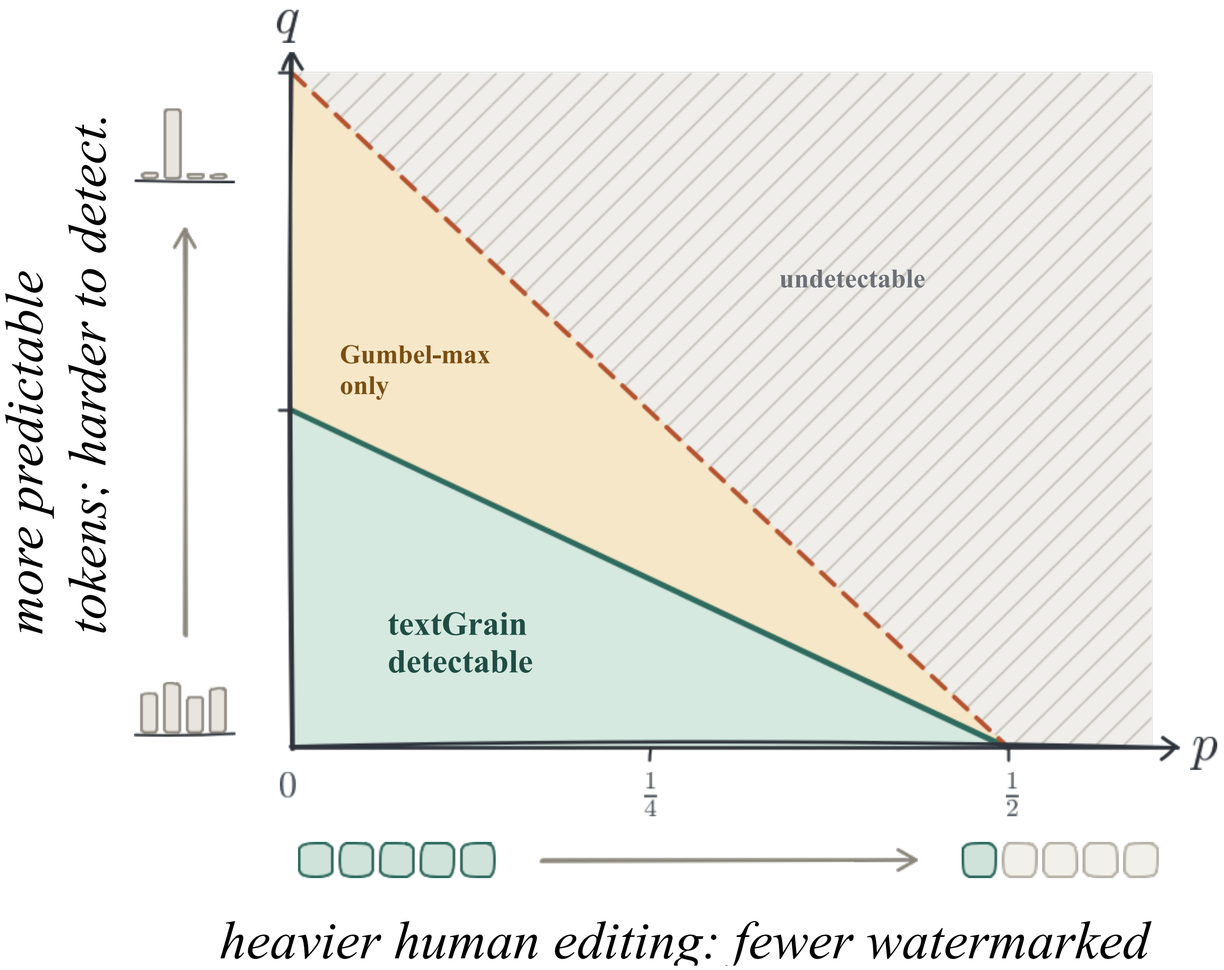
e.g., “2+2=4” irrespective of watermarking



heavier human editing: fewer watermarked tokens

Detection boundaries for textGrain

Bhattacharyya and **B.** (2026)



$\varepsilon_n \asymp n^{-p}$ *share of tokens still watermarked*

$\Delta_n \asymp n^{-q}$ *uncertainty of the next token*

$p + q = \frac{1}{2}$

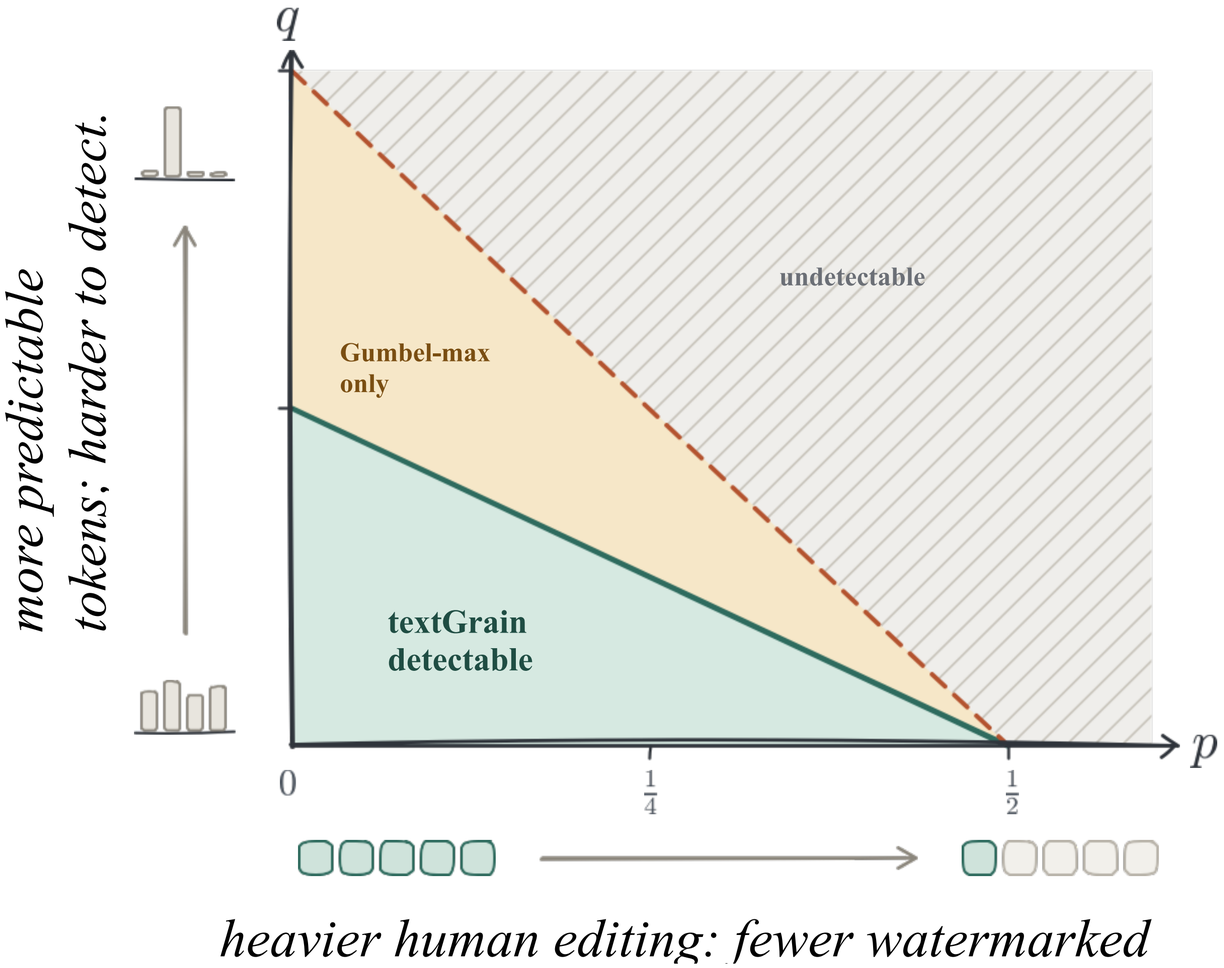
textGrain boundary

$2p + q = 1$

Gumbel-max and maximal-coupling boundary

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textGrain boundary

$2p + q = 1$

Gumbel-max and maximal-coupling boundary

Calibrating entropy costs detectability.

Takeaways

- Watermark localization is really a (interval) change-point problem
  (Much like watermark detection is a two-sample testing/ testing of independence problem)

- The dependency structure is non-stationary, and requires careful segmentation and pruning.

- WISER is fast (linear time if selected block size is $\asymp \sqrt{n}$), theoretically consistent, and almost matches oracle performance in real data.

- textGrain controls how far in terms of entropy you can be from unwatermarked data; this makes detecion harder.

Thank you. Questions?

Paper 1 (WISER)

ArXiv: https://arxiv.org/abs/2509.21160

Code : https://github.com/subroy13/WISER

Paper 2 (textGrain detection boundary)

ArXiv: https://arxiv.org/pdf/2610.17655